
Is pLDDT < 50 a Disorder Detector?

An Operating-Point Audit of AlphaFold2 Confidence Against Experimental Disorder Annotations

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Abstract

AlphaFold DB covers more than 200 million proteins, and the rule “pLDDT < 50 means intrinsically disordered” has become an everyday working heuristic for trimming models and annotating the dark proteome. The rule has never been validated at the operating point at which it is used: the literature reports ROC-AUC, a threshold-free summary that cannot adjudicate a claim about a specific cutoff. We audit the rule directly on the CAID round-2 and round-3 blind assessments, across both official reference conventions (DISORDER-PDB and DISORDER-NOX), with bootstrap confidence intervals over targets and coverage-matched target sets, against a panel of 37–56 dedicated disorder predictors. We find that pLDDT < 50 has sensitivity 0.60–0.63 (0.63–0.65 restricted to regions longer than 30 residues), not above 0.8: not one of 1000 bootstrap resamples reaches 0.80 in any of four settings. Its specificity is 0.98 under DISORDER-PDB and 0.67–0.76 under DISORDER-NOX— a 30-point swing produced entirely by how the benchmark defines a negative, not by AlphaFold. Once every method is threshold-calibrated, none of 174 method $\times$ dataset comparisons beats pLDDT on short (<20 aa) segments, and the median dedicated predictor is 1.5–8.7 points *worse*; the widely assumed 10–15% advantage is exactly reproducible at a shared naive 0.5 cut and vanishes after calibration. What survives is the mechanism: reading solvent accessibility out of the *same* AlphaFold2 coordinates is significantly better on every reference set (Δ AUC +0.015 to +0.056, all $p \leq 0.02$), and the disorder pLDDT misses is confidently modelled and solvent-exposed. pLDDT < 50 is a high-precision, low-recall detector; users who need recall should raise the cutoff to ≈ 70 and prefer the geometric readout of the same model.

1 Introduction

AlphaFold2 attaches a per-residue confidence score, pLDDT, to every model it builds [Jumper et al., 2021]. Almost immediately after the human proteome was released, it was noticed that regions with low pLDDT line up strikingly well with known intrinsically disordered regions (IDRs) [Tunyasuvunakool et al., 2021]. That observation hardened into a rule of thumb: *pLDDT < 50 means disordered*. The rule is now used every day to trim models, to annotate the dark proteome, and to nominate IDR candidates for experiment, across a database of more than 200 million structures [Varadi et al., 2022]. If it is miscalibrated, the error propagates silently into every study that adopts it.

The rule has never been checked where it is used. Prior evaluations of pLDDT as a disorder signal report ROC-AUC [Piovesan et al., 2022, Kurgan et al., 2023]. AUC is threshold-free, and a threshold-free summary is structurally incapable of adjudicating a claim about a specific cutoff. Nobody has reported the sensitivity and specificity of the literal pLDDT < 50 cut, stratified by region length, on both of the reference conventions that the community’s own blind assessment publishes. That is the gap we fill.

What exactly is being claimed? The folk rule comes bundled with a set of quantitative expectations that are worth writing down and testing separately. We decompose the standard framing into four claims: **C1** the rule finds long (>30 aa) IDRs with sensitivity above 0.8; **C2** its specificity is below 0.7; **C3** it works this way because PLDDT conflates conformational flexibility with prediction uncertainty; and **C4** dedicated disorder predictors achieve 10–15% better balanced accuracy on short (<20 aa) disordered segments. Each clause implies a different remedy, so each deserves its own measurement.

Our approach. We do not train or run a new predictor. The CAID blind assessments [Del Conte et al., 2023, Mehdiabadi et al., 2026] already contain the experimental ground truth (derived from DISPROT [Quaglia et al., 2022]) and the per-residue outputs of every participating method, *including the official AlphaFold-pLDDT baseline that the rule refers to*. We evaluate that baseline at the exact operating point in question, on all four combinations of CAID round (2, 3) and reference convention (DISORDER-PDB, DISORDER-NOX), with bootstrap confidence intervals resampled over targets, coverage-matched target sets, and three distinct threshold conventions for every comparator (figure 1). We additionally re-parse 588 ALPHAFOLD DB models to characterise, structurally, the residues the rule gets wrong.

What we find. Three of the four clauses are false and one is right. Sensitivity is 0.60–0.63, and 0.63–0.65 on long regions — $P(\text{sens} \geq 0.80) = 0.000$ in every setting (**C1** falsified). Specificity is 0.98 under DISORDER-PDB and 0.67–0.76 under DISORDER-NOX, so **C2** describes an annotation convention rather than a predictor. After fair threshold calibration, *zero* of 174 method $\times$ dataset comparisons beat PLDDT on short segments and the median comparator is 1.5–8.7 percentage points worse, yet the very same claim is exactly reproducible at a shared naive 0.5 cut, where 9 of 52 predictors land inside the hypothesised +10 to +15 pp band (**C4** falsified, and diagnosed as a calibration artefact). The mechanism clause survives four independent tests: solvent accessibility read from the same coordinates beats PLDDT by +0.015 to +0.056 AUC ($p \leq 0.02$ everywhere), a stronger folding model does not improve PLDDT, PLDDT is near chance *within* its own low band (AUC 0.44–0.63) where geometry still separates disorder at 0.56–0.83, and the disorder the rule misses is confidently modelled (mean PLDDT 70–74) and solvent-exposed (mean RSA 0.46–0.51).

Contributions.

- We report the first operating-point-resolved evaluation of the literal $\text{PLDDT} < 50$ rule — sensitivity, specificity, balanced accuracy and MCC with bootstrap CIs — on all four CAID round $\times$ reference-convention combinations, and show the rule is high-precision and low-recall, missing $\approx 40\%$ of annotated disorder.
- We stratify *regions* (not proteins) into $<20 / 20\text{--}30 / >30$ aa bins under two explicitly stated definitions of “balanced accuracy on short segments”, and show that no dedicated predictor significantly beats PLDDT there once thresholds are calibrated.
- We diagnose the widely believed short-segment advantage as a threshold-calibration artefact, reproducing it exactly at a shared 0.5 cut and dissolving it under calibration — a methodological warning that generalises beyond disorder prediction.
- We test the conflation mechanism with the structure model held fixed, comparing PLDDT against solvent accessibility of the same AlphaFold2 coordinates, and characterise the missed disorder as exposed pseudostructure rather than conditional folding.

section 2 places the work against prior evaluations, section 3 describes the data, panel, operating points and statistics, section 4 reports the four clause tests, and section 5 discusses limitations and practical consequences.

2 Related Work

AlphaFold confidence as a disorder signal. Jumper et al. [2021] introduced PLDDT as a predicted IDDT- $C\alpha$ against the unknown true structure, and Tunyasuvunakool et al. [2021] reported the correlation between low-confidence regions and known IDRs at proteome scale. Piovesan et al. [2022] turned that correlation into concrete, evaluable baselines: ALPHAFOLD-PLDDT with score $= 1 - \text{PLDDT}/100$, and ALPHAFOLD-RSA, the DSSP relative solvent accessibility of the same model smoothed over a 25-residue window. These are the definitions CAID has used ever since, and the first of them is exactly the object of the folk rule — score > 0.50 is identically $\text{PLDDT} < 50$.

Wilson et al. [2022] asked early whether PLDDT should be used as a disorder predictor at all and cautioned that a real correlation does not make the semantics the same. All of this work reports AUC. Unlike these studies, we evaluate the baseline at the specific cutoff practitioners actually apply.

Blind assessment of disorder predictors. CAID [Necci et al., 2021] established the evaluation protocol we inherit: DISPROT-derived references [Quaglia et al., 2022], per-residue scores, blind submission, bootstrap over targets. Round 2 [Del Conte et al., 2023] was the first to include the AlphaFold baselines as official entries; round 3 [Mehdiabadi et al., 2026] added 24 methods, reported that protein language models now lead, and noted that ALPHAFOLD-PLDDT ranks 11th while ALPHAFOLD3-PLDDT ranks 13th on DISORDER-PDB. CAID publishes two disorder references that differ *only* in how a negative is defined: DISORDER-PDB counts as negative only residues observed in an experimental PDB structure, whereas DISORDER-NOX counts every non-annotated residue as negative. CAID3 quantifies the divergence (57.3% vs 23.9% positive content on shared proteins) but treats it as a caveat. We treat it as the dominant term in specificity and report every result under both conventions.

The closest prior comparison. Kurgan et al. [2023] is the nearest work to ours: 20 comparators on CAID1, AF2-PLDDT at AUC 0.722 against fIDPnn [Hu et al., 2021] at 0.814 and AF2-RSA at 0.768, with a long (>30) versus short (≤ 30) IDR split. Two features of that design leave our questions open. Their split is at the *protein* level — proteins whose IDRs are all short — rather than at the region level, so the <20 aa region bin named in the folk rule is unexamined; and their metric is AUC, so no operating point is adjudicated. They also state the conflation mechanism informally, observing that low PLDDT could mean the structure space is poorly covered *or* that the segment is disordered, while unusually high solvent accessibility “seems to be a better proxy” for disorder. Building on that observation, we turn it into a controlled test: same coordinates, two readouts, paired bootstrap. We also adopt their content-matched calibration as one of our three operating points.

What low confidence actually means. Richardson et al. [2025] partition low-PLDDT residues into barbed wire, pseudostructure, near-predictive and unphysical modes, and find that the barbed-wire and near-predictive curves cross near PLDDT 50 — so a cut at 50 cleanly removes non-predictive residues but cannot separate pseudostructure. Symmetrically, Alderson et al. [2023] show that IDRs which conditionally fold are modelled with *high* PLDDT. Both halves of the conflation are therefore documented in the literature; what is missing is a measurement of how much of the rule’s error each half accounts for. We supply that decomposition. Supporting mechanistic work reports that PLDDT tracks dynamics rather than model error alone [Guo et al., 2022], and that AlphaFold2 collapses conformational heterogeneity to a single state [Chakravarty and Porter, 2022, Bhattacharya et al., 2022].

Threshold calibration in method comparison. Kurgan et al. [2023] explicitly calibrate binary predictions to reproduce the correct number of disordered residues, and CAID reports threshold-free metrics precisely because submitted scores are not mutually calibrated. Our **C4** analysis shows what happens when this discipline is dropped: the comparison at a shared 0.5 cut is not a comparison of methods but of operating points, and it produces a specific, reproducible, and wrong quantitative conclusion.

3 Methodology

3.1 Problem statement and design

We test an evaluation claim about a fixed operating point, so the right instrument is an existing blind benchmark rather than a new prediction run. Let $s_m(r) \in [0, 1]$ be method m ’s disorder score for residue r and $y(r) \in \{0, 1\}$ the experimental label. The rule under audit is the classifier $\hat{y}(r) = \mathbb{I}[\text{pLDDT}(r) < 50]$, which under the CAID encoding $s_{\text{ALPHAFOLD-PLDDT}} = 1 - \text{PLDDT}/100$ [Piovesan et al., 2022] is identically $\mathbb{I}[s(r) > 0.5]$. Nothing has to be re-predicted: the CAID archives contain both the ground truth and the official ALPHAFOLD-PLDDT submission. We downloaded AlphaFold2 coordinates separately only to characterise the errors structurally and to verify that the CAID scores really are $1 - \text{PLDDT}/100$.

Four design decisions each answer a specific way the result could have been an artefact. (i) Specificity depends entirely on how negatives are defined, so both CAID references are reported for every result and neither is treated as “the” answer. (ii) AlphaFold covers only 82–86% of CAID2 targets

Reference set	Targets	Positives	Negatives	Excluded	Release
CAID2 DISORDER-PDB	348	37,072	93,805	156,143	CAID2
CAID2 DISORDER-NOX	210	31,315	129,487	0	CAID2
CAID3 DISORDER-PDB	319	31,401	67,838	61,183	CAID3 v3
CAID3 DISORDER-NOX	204	26,367	73,610	541	CAID3 v3

Table 1: The four evaluation references. Residue counts are as published by CAID. The “Excluded” column holds residues with no evidence either way; DISORDER-PDB discards them, DISORDER-NOX treats almost all of them as ordered. This single convention difference drives the specificity result in section 4.2.

while other methods cover $\approx 100\%$, so every comparison runs on the intersection of targets covered by all panel methods; the **C1/C2** tests are additionally repeated on PLDDT’s own coverage and on the full reference with missing targets scored as all-ordered. (iii) Comparing all methods at a shared 0.5 cut is meaningful only for PLDDT, so every method is evaluated at three operating points. (iv) Choosing a threshold on the data one then scores is optimistically biased, so the optimism is measured directly by repeated cross-validation and reported.

3.2 Data

We use the four CAID disorder reference sets (table 1). The two conventions differ only in the definition of a negative: DISORDER-PDB counts as negative only residues observed in an experimental PDB structure, excluding X-ray-missing residues (hence the large excluded column); DISORDER-NOX counts every non-annotated residue as negative. We identified which CAID release each downloaded reference corresponds to by exact matching on (#sequences, positives, negatives, undefined) against the four official reference-set manifests; CAID3 turns out to be the v3 round, so all official comparisons use the v3 files.

We additionally re-parsed 588 ALPHAFOLD DB models (94% of mappable CAID targets) into per-residue PLDDT and Shrake–Rupley relative solvent accessibility for the structural error analysis.

3.3 Comparator panel and operating points

Panel. We restrict the panel to *general* intrinsic-disorder predictors: 37 methods on CAID2 and 56 on CAID3. It spans classical predictors (IUPred3 [Erdős et al., 2021], ESpritz-D [Walsh et al., 2012], MobiDB-lite [Necci et al., 2017, Piovesan et al., 2025], VSL2, RONN, DisEMBL), strong sequence-based methods (fDPnn [Hu et al., 2021], AIUPred, DISOPRED3 [Jones and Cozzetto, 2015], SPOT-Disorder2 [Hanson et al., 2019], Metapredict [Emenecker et al., 2021]) and protein-language-model methods (PUNCH2, SETH [Ilzhöfer et al., 2022], DisorderUnetLM, ESMDisPred, Dispredict3). Methods that predict a different target — disordered *binding* regions (ANCHOR2, MoRFchibi, DisorDPbind, ...) or linkers — are excluded, because including them would flatter PLDDT. Two entries (FoldUnfold, NeProc-disorder) submitted binary states with an empty score column and cannot be threshold-calibrated; they are excluded and the exclusion is logged. Excluding them is the conservative choice, since at their implicit 0.5 cut they would have added two artificially weak comparators to the **C4** test. AlphaFold-derived entries (ALPHAFOLD-PLDDT, ALPHAFOLD-RSA, ALPHAFOLD3-PLDDT, ALPHAFOLD3-RSA) are tracked separately as the object of study and its mechanistic controls, leaving 35 and 52 *dedicated* predictors respectively.

Three operating points. Every method is scored at (a) the **literal** threshold 0.5, which is the only point at which the folk rule is defined; (b) a **calibrated** threshold maximising MCC on the reference, which is our primary setting; and (c) a **content-matched** threshold reproducing the true number of disordered residues, following Kurgan et al. [2023]. The literal comparison is reported precisely to show how misleading it is.

3.4 Metrics and statistics

We compute residue-level sensitivity, specificity, balanced accuracy (BACC), MCC, precision, F1 and ROC-AUC, pooled over residues following CAID’s “dataset” convention. Uncertainty comes from bootstrap resampling *over targets* (1000 resamples for binary metrics, 500 for AUC), which is

CAID’s own resampling unit and respects the fact that residues within a protein are not independent. All method-vs-PLDDT comparisons use a paired bootstrap with identical target indices for both methods, so the interval is on the difference, with a two-sided bootstrap p -value. Clause tests **C1** and **C2** are one-sided bootstrap probabilities, e.g., $P(\text{sensitivity} \geq 0.80)$ over resamples.

Two definitions of “balanced accuracy on short segments”. The folk claim does not define this quantity, so we compute two and report both. The **global-negative** (strict) definition pairs recall over positive residues in 1–19 aa regions with specificity over *all* evaluated negatives, so a method cannot buy short-region recall with false positives elsewhere. The **locally-flanked** (generous) definition scores each region only against the ordered residues immediately flanking it (up to $\lceil L/2 \rceil$ evaluated negatives per side), which removes the influence of a method’s global false-positive rate entirely. We also report region-level detection rates — a region counts as detected when $\geq 25/50/75\%$ of its evaluated residues are predicted disordered — since the claim is phrased about regions.

3.5 Pipeline validation

Before running any experiment we validated the pipeline against the official CAID outputs. For every full-coverage method our pooled AUC reproduces the published value to three decimals on all four references (e.g., IUPred3 0.8854 vs 0.885, ESpritz-D 0.8989 vs 0.899, MobiDB-lite 0.8680 vs 0.868, SETH-1 0.9115 vs 0.911 on CAID2 DISORDER-PDB; fIDPnn 0.8609 vs 0.861 on CAID3 DISORDER-PDB). One trap is worth flagging: the official timing file is keyed by software *group* and reports the group’s best member, so the ALPHAFOLD-DISORDER group value of 0.947–0.95 is ALPHAFOLD-RSA (ours 0.9443/0.9498), not ALPHAFOLD-PLDDT (ours 0.9287/0.9342). Our threshold sweep matches the official per-threshold specificity to three decimals at every probed threshold. Finally, parsing PLDDT out of the ALPHAFOLD DB coordinates and applying $1 - \text{PLDDT}/100$ reproduces the CAID scores at pooled Pearson $r = 0.9925$ over 323,327 residues in 567 targets (median per-target $r = 0.9955$; median residual 1.14 PLDDT units), confirming that the submitted baseline is the standard PLDDT. The residual is ALPHAFOLD DB release drift between the CAID submission and our 2026 snapshot; all benchmark results use the official CAID prediction files, so it does not propagate.

3.6 Reproducibility

Python 3.12.8 with numpy 2.5.3, scipy 1.18.1, pandas 3.0.5, scikit-learn 1.9.0, biopython 1.88. Seed 42 throughout; all bootstrap index matrices are seeded and shared across methods so that paired comparisons use identical resamples. No model is trained or run — all predictions were downloaded — so the analysis is CPU-only and completes in under 20 minutes of wall clock. Re-running the length-stratified and mechanism experiments end to end produced byte-identical output files.

4 Results

4.1 C1: sensitivity at PLDDT < 50 is 0.6, not 0.8

table 2 gives the headline numbers. Sensitivity of the literal rule is 0.599–0.629 over all disordered residues and 0.630–0.648 restricted to regions longer than 30 aa. Not one bootstrap resample out of 1000 reached sensitivity 0.80, in any of the four settings, under either measurement: $P(\text{sens} \geq 0.80) = 0.000$ throughout. The clause fails by roughly 0.17 in absolute terms, an order of magnitude larger than the CI half-width.

The result is not an artefact of the target set. Sensitivity ranges over 0.51–0.63 across three target-set conventions — the full-panel intersection, PLDDT’s own coverage, and the full reference with AlphaFold’s 49 missing CAID2 targets scored as all-ordered (table 9 in the appendix). The pessimistic convention lowers sensitivity further; nothing raises it near 0.8.

Where does sensitivity 0.80 live? Figure 2 answers this directly. Sweeping the cutoff, sensitivity first reaches 0.80 at PLDDT < 75 on both CAID2 references, PLDDT < 74 on CAID3 DISORDER-PDB and PLDDT < 68 on CAID3 DISORDER-NOX. The specificity paid there is 0.94 on DISORDER-PDB — still excellent — and 0.60–0.76 on DISORDER-NOX. This is the study’s most actionable single number: **the rule is set roughly 20 PLDDT units too low.** The MCC-

Clauses C1 and C2: AlphaFold2 pLDDT < 50 against experimental disorder

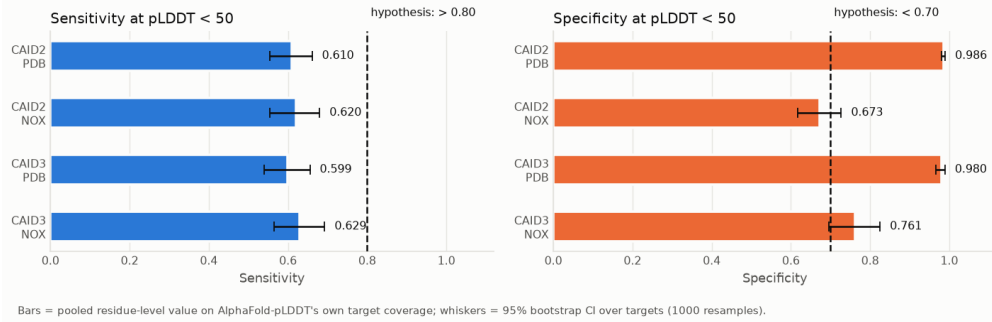


Figure 1: Verdicts on the four clauses. Sensitivity of PLDDT<50 sits at 0.60–0.63 on every reference set, far below the claimed 0.8 (dashed line), and restricting to regions longer than 30 aa moves it only to 0.63–0.65. Specificity, at the identical operating point, is 0.98 under DISORDER-PDB and 0.67–0.76 under DISORDER-NOX: the claim “specificity < 0.7” is a property of the negative-set convention, not of AlphaFold. Error bars are 95% bootstrap intervals over targets.

Reference set	<i>n</i>	Content	Sensitivity (95% CI)	Sens. >30 aa	Specificity	<i>P</i> (sens ≥ 0.8)
CAID2 DISORDER-PDB	299	0.299	0.610 [0.554, 0.661]	0.634 [0.573, 0.687]	0.986	0.000
CAID2 DISORDER-NOX	173	0.235	0.620 [0.553, 0.680]	0.630 [0.562, 0.693]	0.673	0.000
CAID3 DISORDER-PDB	319	0.316	0.599 [0.540, 0.655]	0.630 [0.564, 0.691]	0.980	0.000
CAID3 DISORDER-NOX	204	0.264	0.629 [0.565, 0.693]	0.648 [0.580, 0.711]	0.761	0.000

Table 2: Clause tests for PLDDT<50 on PLDDT’s own target coverage. “Content” is the fraction of evaluated residues that are disordered. Specificity 95% CIs are [0.981, 0.990], [0.618, 0.727], [0.966, 0.989] and [0.696, 0.825] respectively. Sensitivity never approaches 0.8; specificity is decided by the reference convention.

optimal disorder cutoff on this data is PLDDT $\approx$ 69–72, and at PLDDT < 70 sensitivity is 0.77–0.82 at specificity 0.95 on DISORDER-PDB.

4.2 C2: specificity is decided by the negative-set convention

At the identical operating point, on the same residues, with the same predictor, specificity is 0.98 under DISORDER-PDB and 0.67–0.76 under DISORDER-NOX. The claim “specificity < 0.7” is decisively false on both DISORDER-PDB sets ($P(\text{spec} \leq 0.70) = 0.000$), consistent with the data on CAID2 DISORDER-NOX (0.673, $P = 0.815$), and false on CAID3 DISORDER-NOX (0.761, $P = 0.032$, i.e., significantly *above* 0.7).

The empirical calibration in table 3 shows the mechanism. Under DISORDER-PDB, residues in X-ray-missing segments are excluded from the negative set, so almost everything left below PLDDT 50 is a true positive and precision is 0.93–0.95. Under DISORDER-NOX those same residues become negatives, and precision collapses to 0.37–0.49 with no change whatsoever in the predictor. Note the invariant in the last column: **PLDDT<50 captures 60–63% of all annotated disorder regardless of convention**. That is the stable, method-intrinsic quantity; what changes is what the complement is called. Recall is the method’s property, precision is the benchmark’s.

4.3 C4: the short-segment gap is real inside PLDDT, but dedicated predictors do not close it

PLDDT does degrade on short regions. table 4 shows recall falling from 0.58–0.61 on regions longer than 30 aa to 0.25–0.43 on regions shorter than 20 aa at the literal cut. Calibration lifts every stratum but preserves the ordering (0.59–0.65 short versus 0.78–0.85 long). Region-level detection at 50% overlap tells the same story: 59–81% of short regions detected versus 80–89% of long ones (table 10, appendix). So the premise behind **C4**— short disorder is harder for PLDDT— is correct.

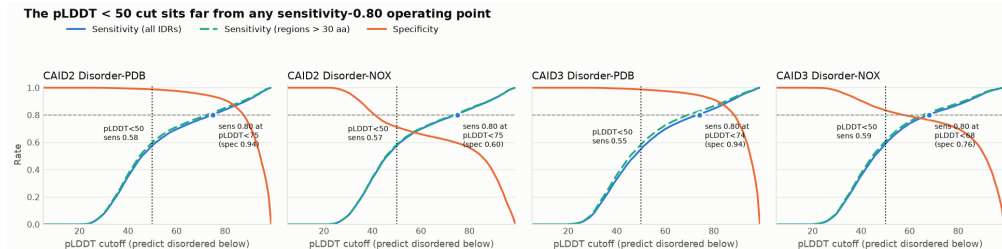


Figure 2: Sensitivity and specificity of a PLDDT cutoff as the cutoff is swept from 5 to 99. The vertical line marks the folk rule at 50. The AlphaFold “very low confidence” band was never calibrated against disorder annotations: on all four references the MCC-optimal disorder cutoff falls at PLDDT 69–72, and sensitivity 0.80 is not reached until PLDDT 68–75.

Reference set	Prevalence	% res. PLDDT <50	$P(\text{dis.} \text{PLDDT} < 50)$	$P(\text{dis.} \text{PLDDT} \geq 50)$	Disorder below 50
CAID2 DISORDER-PDB	0.299	19.3	0.948	0.145	0.610
CAID2 DISORDER-NOX	0.235	39.6	0.368	0.148	0.620
CAID3 DISORDER-PDB	0.316	20.4	0.932	0.159	0.599
CAID3 DISORDER-NOX	0.264	34.1	0.486	0.149	0.629

Table 3: Empirical calibration of the rule. The precision of $\text{PLDDT} < 50$ swings from 0.95 to 0.37 purely with the negative-set convention, while the share of all annotated disorder that falls below PLDDT 50 is invariant at 0.60–0.63. Reporting one convention and calling its specificity “the” specificity is the single most consequential methodological choice in this literature.

But the comparative claim fails. table 5 gives the verdict. With every method calibrated to its own MCC-optimal threshold, *zero* of the 174 method $\times$ dataset comparisons is significantly better than PLDDT on short segments, and the median dedicated predictor is 1.5–8.7 pp *worse*. The best single result anywhere is +6.8 pp (fIDPnn3a on CAID3 DISORDER-NOX, CI $[-0.9, +14.5]$, not significant). Under the deliberately generous locally-flanked definition, which removes a method’s global false-positive rate from the picture entirely, the gap gets *wider* rather than narrower (−4.7 to −11.2 pp). The interaction test — does the dedicated-predictor advantage grow specifically on short regions, which is the substantive content of the claim — is negative on three of four datasets (median −5.1, −4.9, −1.4 pp, and +1.7 pp on CAID3 DISORDER-NOX), with 19, 14, 17 and 7 methods showing a significantly *smaller* advantage on short regions than on long ones.

Why the artefact arises. At 0.5, PLDDT sits at specificity 0.99 and recall 0.33 — an extremely conservative corner of its own ROC curve — while a permissive predictor sits near specificity 0.65 and recall 0.70. That is a comparison of operating points, not of methods. Calibration moves fIDPnn3a from +19.3 pp to +6.8 pp and UdonPred-combined from +13.5 pp to below the median. The folk claim is a real, reproducible measurement of an artefact (figure 3).

4.4 Where PLDDT actually stands, fairly compared

Falsifying the folk claims cuts both ways. Calibrated and coverage-matched on the common target set, PLDDT is a respectable mid-pack disorder predictor (table 6): rank 9/37 and 11/56 on DISORDER-PDB, 16/37 and 17/56 on DISORDER-NOX, comparable to IUPred3 and DISOPRED3, behind PUNCH2, SETH-0 and the PredIDR family, ahead of fIDPnn, ESpritz-D and MobiDB-lite. That is a considerably more favourable verdict than the folk framing implies, and a considerably less favourable one than “PLDDT < 50 finds disorder” implies. Calibration optimism is negligible: the median gap between in-sample MCC-optimal BACC and the 5×2 -fold cross-validated estimate is 0.20–0.66 pp, with a maximum of 3.0 pp across 186 method $\times$ dataset combinations, so the calibrated rankings are not an artefact of tuning on the evaluation data.

4.5 C3: PLDDT is a lossy readout of a structure that already knows better

Four independent tests support the conflation mechanism.

Reference set	Operating point	Thr.	Spec.	Recall <20 aa	Recall 20–30 aa	Recall >30 aa
CAID2 DISORDER-PDB	literal	0.500	0.988	0.393 [0.324, 0.457]	0.485 [0.379, 0.574]	0.598 [0.537, 0.654]
	calibrated	0.308	0.956	0.653 [0.579, 0.721]	0.691 [0.600, 0.777]	0.779 [0.727, 0.827]
CAID2 DISORDER-NOX	literal	0.500	0.712	0.433 [0.284, 0.570]	0.445 [0.307, 0.567]	0.583 [0.521, 0.651]
	calibrated	0.206	0.566	0.806 [0.694, 0.912]	0.726 [0.571, 0.861]	0.844 [0.793, 0.891]
CAID3 DISORDER-PDB	literal	0.500	0.987	0.331 [0.265, 0.397]	0.407 [0.316, 0.492]	0.583 [0.525, 0.637]
	calibrated	0.282	0.952	0.628 [0.555, 0.699]	0.682 [0.585, 0.776]	0.816 [0.769, 0.857]
CAID3 DISORDER-NOX	literal	0.500	0.835	0.250 [0.158, 0.352]	0.428 [0.306, 0.558]	0.608 [0.554, 0.660]
	calibrated	0.272	0.745	0.587 [0.465, 0.710]	0.726 [0.596, 0.852]	0.846 [0.808, 0.881]

Table 4: Region-length-stratified recall of pLDDT with 95% bootstrap CIs over targets. Short regions are genuinely harder, at both operating points. Calibration raises recall everywhere by 20–35 points at a cost of 3–17 points of specificity, which is why comparing methods at a shared threshold is not a comparison of methods.

Reference set	n	Calibrated thresholds (primary)					Shared naive 0.5 cut		
		pLDDT BACC	Δ med.	Δ best	sig. better	in band	Δ med.	sig. better	in band
CAID2 DISORDER-PDB	35	0.804	−8.7	+0.1	0	0	−0.2	7	3
CAID2 DISORDER-NOX	35	0.686	−7.7	+0.3	0	0	+0.6	3	1
CAID3 DISORDER-PDB	52	0.790	−6.5	+2.4	0	0	+1.9	21	6
CAID3 DISORDER-NOX	52	0.666	−1.5	+6.8	0	0	+7.6	34	9

Table 5: The claimed 10–15% short-segment advantage of dedicated predictors is a calibration artefact. All deltas are percentage points of balanced accuracy on <20 aa regions relative to pLDDT; “in band” counts methods landing inside the claimed +10 to +15 pp window. Left: with every method at its own MCC-optimal threshold, no method is significantly better and the median is worse. Right: cutting every method at 0.5 — the comparison one gets by thresholding published scores — reproduces the claim almost exactly on CAID3 DISORDER-NOX, where 9 of 52 predictors fall inside the band and 34 are significantly better.

Test 1: same model, different readout. ALPHAFOLD-RSA is the solvent accessibility of the *identical* AlphaFold2 coordinates, smoothed over a window. It beats pLDDT on every reference set: Δ AUC +0.015 to +0.056 with $p \leq 0.02$ throughout, and Δ BACC +0.017 to +0.044 (table 7). This is the cleanest available test of the conflation claim, because the structure model is held exactly fixed and only the quantity read out of it changes. The confidence channel discards disorder-relevant information that is present in the coordinates it is attached to.

Test 2: a better folding model does not help. AlphaFold3’s pLDDT is statistically indistinguishable from AlphaFold2’s on both CAID3 references ($p = 0.62$ and 0.61), while AlphaFold3’s RSA is significantly better (+0.059 AUC on DISORDER-NOX, $p = 0.002$). If low pLDDT tracked genuine flexibility, a stronger folding model should sharpen it. It does not: the gain lives in the readout, not in the model.

Test 3: residual signal inside pLDDT strata. Within the pLDDT < 50 stratum, pLDDT itself discriminates disorder at AUC 0.44–0.63 — at or below chance on the DISORDER-NOX references — while RSA of the *same* residues discriminates at 0.56–0.83. In the 50–70 band the gap is widest (pLDDT 0.56–0.68 versus RSA 0.69–0.90; table 11, appendix). The confidence score saturates: once it has said “not confident”, it stops carrying information about how disordered a residue is, while the structure still does (figure 4).

Test 4: information overlap. A 5-fold grouped-by-target cross-validated logistic probe on the raw features — an information probe, not a proposed predictor — gives CV AUC 0.926/0.689/ 0.931/0.761 for pLDDT alone, 0.944/0.741/0.949/0.827 for RSA alone, and 0.949/0.743/0.956/0.822 for both. Adding pLDDT on top of RSA gains at most +0.007 AUC and *loses* 0.005 on CAID3 DISORDER-NOX; adding RSA on top of pLDDT gains +0.023 to +0.065. pLDDT’s disorder information is very nearly a subset of the geometry’s.

4.6 What the rule misses, structurally

Decomposing the residues into true positives, false negatives, false positives and true negatives (table 8) shows what kind of disorder escapes the rule. The missed disorder is **confidently modelled**

Clause C4: the short-segment gap is real inside pLDDT, but dedicated predictors do not close it

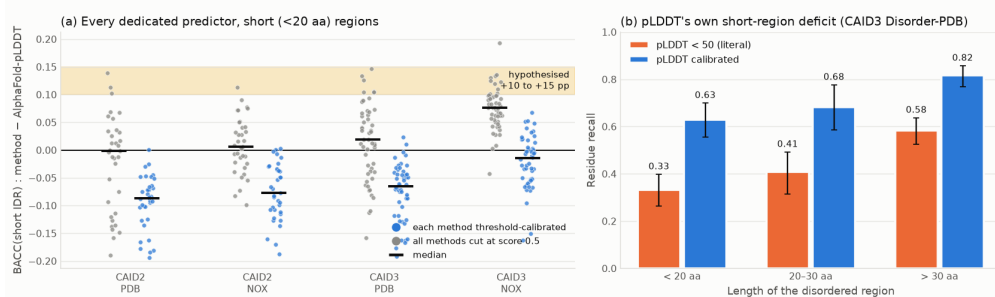


Figure 3: Per-method balanced-accuracy difference from PLDDT on <20 aa disordered regions. The shaded band is the claimed +10 to +15 pp advantage. At calibrated thresholds the distribution sits below zero on every reference set and no method’s CI clears zero; the same methods scored at a shared 0.5 cut populate the band. Both panels use identical residues and identical bootstrap resamples — only the thresholds differ.

Reference set	PLDDT BACC	Rank	PLDDT AUC	Best dedicated method	ALPHA FOLD-RSA BACC
CAID2 DISORDER-PDB	0.860	9 / 37	0.932	PredIDR-long 0.887	0.886
CAID2 DISORDER-NOX	0.703	16 / 37	0.715	SPOT-Disorder2 0.734	0.713
CAID3 DISORDER-PDB	0.871	11 / 56	0.943	PUNCH2 0.887	0.886
CAID3 DISORDER-NOX	0.789	17 / 56	0.833	DisorderUnetLM 0.819	0.803

Table 6: PLDDT at its calibrated operating point against the full panel, coverage-matched. PLDDT is mid-pack, not the outlier the folk rule’s critics or its users assume; and ALPHA FOLD-RSA, derived from the same coordinates at no extra cost, is within a point of the best dedicated method on DISORDER-PDB.

— mean PLDDT 70–74, with 47–56% of those residues above PLDDT 70 and 12–21% above 90 — and **solvent-exposed**, with mean RSA 0.46–0.51 against 0.25–0.31 for true negatives, and only 17–23% buried against 46–57% of true negatives. It sits in shorter regions (median 110–161 aa versus 170–239 for detected disorder) and has an intermediate amino-acid composition: detected disorder shows the classic low-complexity signature ($\log_2$ enrichment S +0.64, P +0.58, G +0.55, with strong hydrophobic depletion I –1.22, W –1.05, F –1.05), whereas missed disorder keeps the charged/proline enrichment (K +0.44, P +0.35, Q +0.27) but has no S/G enrichment (S –0.05, G –0.15) and is only mildly hydrophobic-depleted (I –0.37).

This decomposes the failure into two mechanisms with different remedies. The 17–23% buried fraction is **conditional folding** [Alderson et al., 2023]: AlphaFold confidently builds a folded conformation for a region that is disordered in isolation, and no confidence threshold can recover it. The larger, exposed fraction is **pseudostucture** [Richardson et al., 2025]: AlphaFold confidently models an extended or helical conformation for a region that is genuinely flexible. That fraction *is* recoverable — it is exactly what the RSA readout catches, and it is why RSA beats PLDDT (figure 5).

5 Discussion

5.1 Interpretation

The audited claim is best read as three empirical predictions plus one causal story, and the data separate them cleanly.

The two point predictions are wrong for instructively different reasons. C1 is wrong because the rule is *conservative*, not liberal: it sits roughly 20 PLDDT units below the disorder-optimal operating point and captures only the 19–40% of residues in the lowest confidence band. The framing appears to assume that “very low confidence” is a permissive filter; it is a strict one. C2 is wrong in a deeper way, because it is not a statement about AlphaFold at all. The same predictions, the

Reference set	Comparison	n	Δ AUC	95% CI	p	Δ BACC	p
CAID2 DISORDER-PDB	AF2-RSA – AF2-PLDDT	299	+0.0160	[+0.0056, +0.0256]	0.002	+0.0263	0.001
CAID2 DISORDER-NOX	AF2-RSA – AF2-PLDDT	173	+0.0508	[+0.0201, +0.0894]	0.004	+0.0169	0.032
CAID3 DISORDER-PDB	AF2-RSA – AF2-PLDDT	319	+0.0152	[+0.0032, +0.0311]	0.020	+0.0272	0.001
CAID3 DISORDER-NOX	AF2-RSA – AF2-PLDDT	204	+0.0561	[+0.0180, +0.0980]	0.002	+0.0435	0.002
CAID3 DISORDER-PDB	AF3-PLDDT – AF2-PLDDT	319	−0.0021	[−0.0102, +0.0082]	0.616	+0.0010	0.856
CAID3 DISORDER-NOX	AF3-PLDDT – AF2-PLDDT	204	+0.0072	[−0.0150, +0.0341]	0.612	+0.0106	0.356
CAID3 DISORDER-NOX	AF3-RSA – AF2-PLDDT	204	+0.0585	[+0.0141, +0.1136]	0.002	+0.0513	0.002

Table 7: Controlled readout comparisons by paired bootstrap over targets. Changing the readout of a fixed AlphaFold2 model from confidence to geometry helps significantly everywhere; changing the folding model while keeping the confidence readout does not help at all.

Clause C3: pLDDT is a lossy readout of a structure that already encodes disorder better

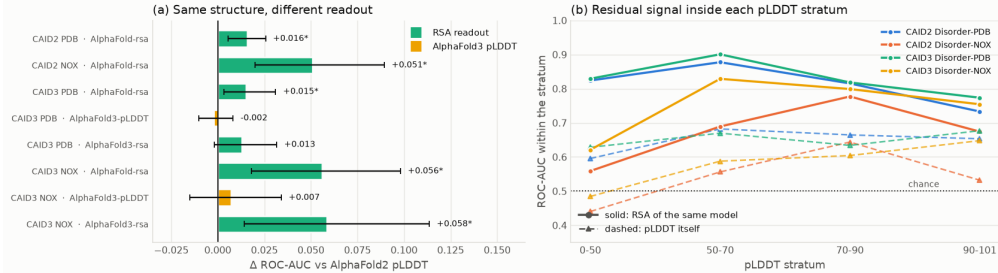


Figure 4: The conflation mechanism. (a) Reading solvent accessibility instead of confidence out of the same AlphaFold2 coordinates improves AUC on all four reference sets, while replacing AlphaFold2 with AlphaFold3 does not. (b) Within PLDDT strata, PLDDT has saturated — inside its own <50 band it discriminates disorder at chance level on the DISORDER-NOX references — while the geometry of the same model still separates disorder from order.

same residues and the same threshold give specificity 0.98 or 0.67 depending on whether the benchmark counts X-ray-missing residues as negatives. A claim whose truth value flips on an annotation convention is under-specified, and the fix is simply to state the convention.

The comparative prediction is wrong in the opposite direction, and its origin is diagnosable. Once thresholds are calibrated, PLDDT is *better* than the median dedicated predictor on short segments, on all four datasets, under both a strict and a generous definition, with zero significant exceptions among 174 comparisons. The claim is nonetheless recoverable at a shared 0.5 threshold, where it is not only true but quantitatively accurate. We think this is how the belief arises, and the lesson generalises: whenever a comparison mixes a method whose scores are physically anchored (PLDDT is a predicted lDDT, so 0.5 means something specific about geometry) with methods whose scores are arbitrary monotone confidences, a shared cut compares operating points rather than methods.

The causal story is right, and more specific than usually stated. Four independent tests support conflation: RSA of the same coordinates beats PLDDT on every reference set; a stronger folding model does not improve PLDDT but does improve RSA; inside the low-PLDDT stratum PLDDT is at chance while RSA is not; and pLDDT adds at most +0.007 AUC on top of RSA while RSA adds up to +0.065 on top of PLDDT. The error-stratum analysis sharpens “conflation” into a concrete failure mode: PLDDT is high wherever AlphaFold commits to a conformation, whether or not that conformation is real. Most missed disorder is confidently modelled and solvent-exposed — extended pseudostructure — which is why a geometric readout rescues it and why raising the confidence threshold only partially helps.

5.2 Comparison with prior work

Our AUCs reproduce the official CAID values to three decimals wherever coverage is complete (section 3.5), so the study is anchored to the published assessment. Relative to Kurgan et al. [2023], who report AF2-PLDDT at AUC 0.722 and AF2-RSA at 0.768 on CAID1, we find higher absolute AUCs on CAID2/CAID3 DISORDER-PDB (0.929–0.934 and 0.944–0.950) but the same ordering

Reference set	Stratum	n residues	Mean pLDDT	Mean RSA	% buried	Median region len.
CAID3 DISORDER-PDB	TP (disordered, pLDDT <50)	18,037	37.0	0.638	2.7	170
	FN (disordered, pLDDT $\geq$50)	12,202	71.4	0.492	19.2	110
	FP (ordered, pLDDT <50)	886	41.4	0.547	13.9	—
	TN (ordered, pLDDT $\geq$ 50)	62,840	91.9	0.255	56.1	—
CAID2 DISORDER-PDB	TP	17,186	37.1	0.646	1.8	214
	FN	10,428	73.5	0.463	23.2	125

Table 8: Structural profile of the rule’s error strata (2026 ALPHAFOLD DB snapshot, Shrake–Rupley RSA). “% buried” is the fraction with RSA < 0.25. The disorder the rule misses is not borderline: it is confidently modelled and mostly solvent-exposed, i.e., extended pseudostructure rather than genuinely ambiguous prediction.

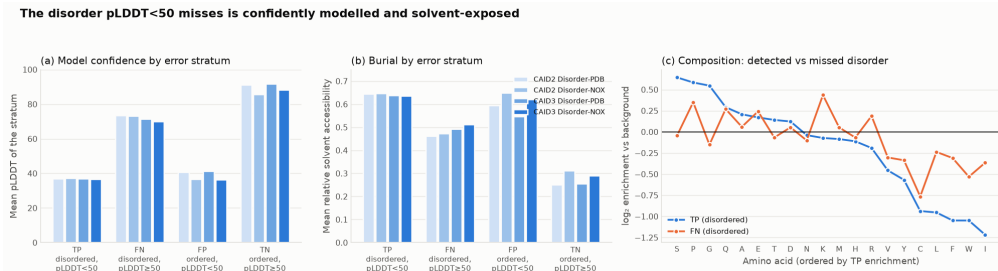


Figure 5: What pLDDT<50 misses. (a)Confidence distribution by stratum: false negatives are not clustered just above the cutoff but spread across the confident range. (b)Solvent accessibility separates false negatives from true negatives even where confidence does not. (c)Amino-acid composition places missed disorder between canonical low-complexity IDRs and ordered sequence, consistent with a conformation AlphaFold can commit to.

and a similar gap. The absolute difference is a reference-set effect: CAID1 differs from CAID2/3, and DISORDER-PDB’s exclusion of X-ray-missing negatives raises all AUCs. Their protein-level finding that short-IDR proteins are harder is reproduced here at region level and shown to be a property of *all* methods rather than a pLDDT-specific weakness. We also confirm the CAID3 observation that ALPHAFOLD3-pLDDT does not improve on ALPHAFOLD-pLDDT, and add that the difference is not statistically distinguishable from zero.

5.3 Practical recommendations

- **Do not use pLDDT<50 as a disorder detector if recall matters.** Use pLDDT < 70 (the MCC-optimal region is 69–72), which raises recall on long regions from ≈ 0.60 to 0.78–0.85 at specificity ≥ 0.95 on DISORDER-PDB.
- **Prefer ALPHAFOLD-RSA to ALPHAFOLD-PLDDT.** It costs nothing extra, since the structure is already computed, and it is significantly better on every reference set tested.
- **State the negative-set convention** whenever reporting specificity or precision for a disorder predictor. Reporting one convention silently determines the answer.
- **Never compare disorder predictors at a shared score threshold.** This paper contains a worked example of the false conclusion that follows.

5.4 Limitations

Methodological. Our primary “calibrated” operating point maximises MCC on the full data, following CAID’s own convention, and is therefore optimistically biased. We measured the optimism by 5×2 -fold cross-validation over targets and found it small (median 0.20–0.66 pp, max 3.0 pp), but it is not zero and is slightly larger on the DISORDER-NOX references where the class balance is less favourable. Balanced accuracy on a length stratum is a constructed quantity that the folk claim does not define; we report two definitions that bracket the reasonable range and agree, but a third could disagree. Our p -values are percentile-bootstrap rather than exact, which can be slightly anti-conservative for small effects; the **C4** conclusion rests on effects of the wrong *sign*, so this is not load-bearing there, but it matters for the marginal AlphaFold3 comparisons, which we report

as null rather than as equivalence. We apply no multiple-comparison correction across the 174 **C4** comparisons; correction would only strengthen the “0 better” conclusion and weaken the “97 worse” one, so we report both counts and lean on the former.

Data and scope. CAID1 is no longer retrievable, so Kurgan et al. [2023] cannot be reproduced exactly on their own data; CAID2 and CAID3 v3 supersede it. The ground truth is DISPROT-derived and inherits DISPROT’s biases: curation is literature-driven and long, well-characterised IDRs are over-represented relative to short ones (1,610–1,808 residues in <20 aa regions versus 16,502–19,703 in >30 aa regions), so short-region estimates are correspondingly noisier, as the wider CIs in table 4 show. Coverage matching costs 18–36% of targets because the intersection is set by the worst-covered panel member; the clause tests are therefore also reported on PLDDT’s own coverage and on the full reference, with unchanged conclusions. Absence of annotation is not annotation of order — DISORDER-NOX assumes it is, DISORDER-PDB refuses to — and neither is a gold standard, which is exactly why both are reported. Our re-parsed ALPHAFOLD DB coordinates differ from the CAID-era models at $r = 0.992$ (median 1.14 PLDDT units), which affects only the descriptive error-stratum statistics, not the benchmark results; likewise the structural analysis uses Shrake–Rupley rather than DSSP accessibility, while the benchmark comparisons use the official DSSP-based ALPHAFOLD-RSA submissions. Finally, the 2-feature logistic probe is linear, so “PLDDT adds ≤ 0.007 on top of RSA” is a statement about linearly accessible information, not an information-theoretic bound. We did not test predicted aligned error, MSA depth, or per-domain analyses.

What could invalidate these results. A ground truth in which short disordered regions are systematically better represented could change the **C4** magnitudes, though we doubt it would change the sign given the size and consistency of the effect. A demonstration that CAID’s ALPHAFOLD-PLDDT submission used a non-standard AlphaFold configuration would undercut the mapping to the folk rule; we checked this as far as the data allow ($r = 0.992$ against ALPHAFOLD DB) and found no evidence of it. And if a reader’s operational definition of “disordered” is “not resolved in a crystal”, then DISORDER-NOX is the wrong reference for them and the specificity story changes — which is itself the paper’s central methodological point.

5.5 Broader implications

Two lessons travel beyond disorder prediction. First, a benchmark that publishes more than one convention for what counts as a negative is publishing more than one question, and a claim stated without naming the convention has no truth value. Second, a folk rule inherited from a model’s internal diagnostic is unlikely to be calibrated for a downstream task it was never trained on: PLDDT targets IDDT- $C\alpha$, not disorder, and the 50 boundary comes from AlphaFold’s own confidence banding, not from any disorder annotation. Both failure modes are easy to reproduce in other settings where a model’s uncertainty output gets repurposed as a semantic label.

6 Conclusion

We audited the rule “AlphaFold2 PLDDT < 50 means intrinsically disordered” at the operating point at which it is actually used, on the CAID round-2 and round-3 blind assessments, under both official reference conventions, against 35–52 dedicated disorder predictors.

Findings. PLDDT <50 does not identify disordered regions with sensitivity above 0.8: it achieves 0.60–0.63 overall and 0.63–0.65 on regions longer than 30 aa, missing roughly 40% of experimentally annotated disorder, with $P(\text{sens} \geq 0.80) = 0.000$ in every setting tested. Its specificity is 0.98 or 0.67–0.76 depending purely on whether the benchmark counts X-ray-missing residues as negatives, so the “specificity < 0.7 ” clause describes an annotation convention rather than a predictor. Dedicated predictors do not achieve 10–15% better balanced accuracy on short segments: after fair threshold calibration, none of them is significantly better and the median is 1.5–8.7 points *worse*; the claimed gap is reproducible only when all methods are cut at a shared, arbitrary threshold. The mechanism, however, is correct and can be demonstrated directly: PLDDT conflates flexibility with model confidence, which is why reading solvent accessibility out of the same AlphaFold2 coordinates is significantly better on every reference set, and why the disorder PLDDT misses is confidently modelled and solvent-exposed.

Takeaway. PLDDT<50 is a high-precision, low-recall disorder detector, and only under the DISORDER-PDB negative-set convention. Raise the cutoff to ≈ 70 if recall matters, prefer the free geometric readout of the same model, and state which negative-set convention you mean.

Future work. Three directions follow directly. The 77–83% of missed disorder that is not buried should be recoverable by a rule combining PLDDT with local RSA and secondary-structure content; quantifying the ceiling of that combination would say how much of the gap is closable without leaving the AlphaFold model. The buried false negatives should be cross-referenced against curated conditionally folded IDR sets [Alderson et al., 2023] to check that they are the same population, and tested against partner-aware predictions. And predicted aligned error is the natural next signal, since a residue can carry high PLDDT locally while being poorly positioned relative to the rest of the chain — which is exactly the pseudostructure signature. Two questions remain open: whether the PLDDT saturation we observe below 50 is intrinsic to the IDDT target or an artefact of how AlphaFold’s confidence head is trained, and whether a benchmark can be built whose negatives are defined by positive experimental evidence of order rather than by absence of evidence of disorder. Both CAID conventions are proxies for a reference set that nobody has built.

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A Robustness to the target-set convention

Clauses **C1** and **C2** concern PLDDT alone and should not be hostage to other panel members’ coverage. table 9 repeats the clause tests under three target-set conventions: the intersection of targets covered by every panel method (used for all method comparisons), PLDDT’s own coverage (used in table 2), and the full reference with targets absent from ALPHAFOLD DB scored as all-ordered. CAID3 has full AlphaFold coverage, so its last two rows coincide. Sensitivity spans 0.51–0.63; the pessimistic convention lowers it, and nothing raises it near 0.8.

Reference set	Target set	<i>n</i>	Sens.	Sens >30	Sens <20	Spec.	BACC	AUC
CAID2 DISORDER-PDB	panel intersection	240	0.576	0.598	0.393	0.988	0.782	0.932
	PLDDT own coverage	299	0.610	0.634	0.416	0.986	0.798	0.929
	all targets, missing = ordered	348	0.511	0.528	0.366	0.989	0.750	0.810
CAID2 DISORDER-NOX	panel intersection	135	0.575	0.583	0.433	0.712	0.644	0.715
	PLDDT own coverage	173	0.620	0.630	0.429	0.673	0.646	0.695
	all targets, missing = ordered	210	0.511	0.520	0.332	0.788	0.649	0.692
CAID3 DISORDER-PDB	panel intersection	261	0.551	0.583	0.331	0.987	0.769	0.943
	PLDDT own coverage / all targets	319	0.599	0.630	0.349	0.980	0.790	0.934
CAID3 DISORDER-NOX	panel intersection	161	0.588	0.608	0.250	0.835	0.712	0.833
	PLDDT own coverage / all targets	204	0.629	0.648	0.264	0.761	0.695	0.778

Table 9: Clause tests under three target-set conventions. The conclusion for **C1** is unchanged in all cases.

B Region-level detection rates

The folk rule is phrased about regions, so table 10 reports region-level detection: a region counts as detected when at least a given fraction of its evaluated residues is predicted disordered at the literal cut. The length ordering of section 4.3 survives at every overlap criterion.

Reference set	Min overlap	Det. <20 aa	<i>n</i>	Det. 20–30 aa	<i>n</i>	Det. >30 aa	<i>n</i>
CAID2 DISORDER-PDB	0.25	0.737	118	0.797	64	0.869	153
	0.50	0.661	118	0.719	64	0.797	153
	0.75	0.576	118	0.625	64	0.641	153
CAID2 DISORDER-NOX	0.25	0.871	31	0.808	26	0.932	117
	0.50	0.806	31	0.769	26	0.863	117
	0.75	0.774	31	0.654	26	0.701	117
CAID3 DISORDER-PDB	0.25	0.772	136	0.797	64	0.907	172
	0.50	0.647	136	0.750	64	0.831	172
	0.75	0.544	136	0.547	64	0.686	172
CAID3 DISORDER-NOX	0.25	0.750	44	0.844	32	0.939	131
	0.50	0.591	44	0.781	32	0.893	131
	0.75	0.523	44	0.625	32	0.740	131

Table 10: Region-level detection rate of PLDDT<50 by region length and overlap criterion.

C Conditional information within PLDDT strata

table 11 reports, within each PLDDT band, how well the solvent accessibility of the same model separates disordered from ordered residues, together with the cross-validated logistic probes of section 4.5. Inside the <50 band — where the confidence signal has saturated and discriminates at AUC 0.44–0.63 — the geometry of the same model still reaches 0.56–0.83.

D Calibration optimism

Our primary operating point selects each method’s MCC-optimal threshold on the evaluation data. table 12 quantifies the resulting optimism by repeated 5×2 -fold cross-validation over targets: the

Reference set	AUC(RSA) within pLDDT band				5-fold CV logistic probe		
	0–50	50–70	70–90	90–100	pLDDT	RSA	both
CAID2 DISORDER-PDB	0.824	0.877	0.815	0.733	0.9263	0.9440	0.9490
CAID2 DISORDER-NOX	0.559	0.689	0.777	0.674	0.6888	0.7405	0.7431
CAID3 DISORDER-PDB	0.829	0.901	0.818	0.773	0.9314	0.9488	0.9555
CAID3 DISORDER-NOX	0.621	0.829	0.799	0.754	0.7613	0.8267	0.8218

Table 11: Residual disorder signal in AlphaFold2 geometry, conditioned on confidence. Adding PLDDT on top of RSA gains at most +0.007 AUC and loses 0.005 on CAID3 DISORDER-NOX, while adding RSA on top of PLDDT gains +0.023 to +0.065.

median gap is 0.20–0.66 percentage points of balanced accuracy, with a maximum of 3.0 pp over 186 method $\times$ dataset combinations. The calibrated rankings in section 4.4 are therefore not an artefact of tuning on the evaluation data.

Reference set	n methods	Median in-sample BACC	Median CV BACC	Median optimism (pp)	Max optimism (pp)
CAID2 DISORDER-PDB	37		0.8175		1.367
CAID2 DISORDER-NOX	37		0.7000		3.036
CAID3 DISORDER-PDB	56		0.8255		1.691
CAID3 DISORDER-NOX	56		0.7571		2.524

Table 12: In-sample versus cross-validated calibrated balanced accuracy.

E Full panel ranking

Figure 6 shows the calibrated, coverage-matched ranking of the whole panel on all four reference sets, with PLDDT and the AlphaFold-derived controls highlighted. Figure 7 shows the empirical probability of disorder as a function of PLDDT bin, which is the graphical form of table 3.

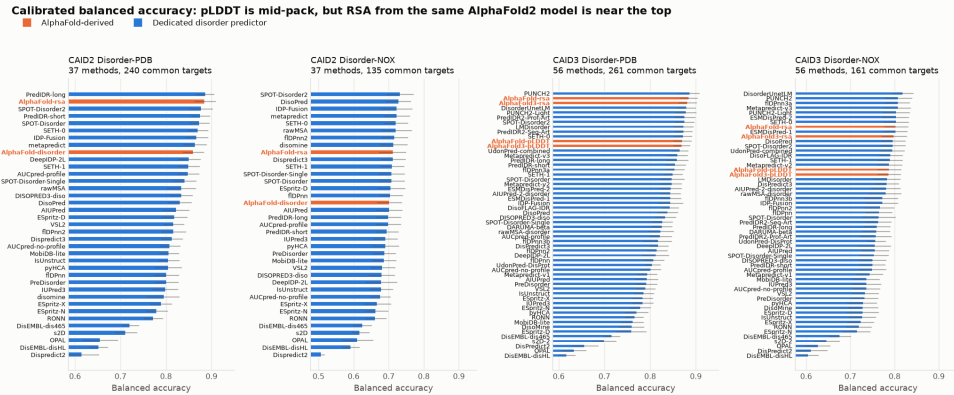


Figure 6: Calibrated balanced accuracy of every panel method on the coverage-matched target sets. PLDDT is mid-pack on all four references; ALPHAFOLD-RSA, read from the same coordinates, is consistently near the top.

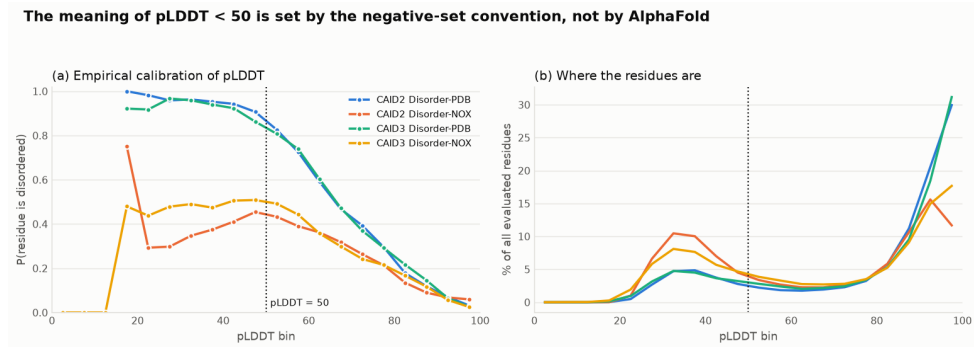


Figure 7: Empirical probability that a residue is annotated disordered, as a function of its pLDDT bin. The curves for the two negative-set conventions are shifted by roughly a factor of two in precision at every bin, while the fraction of all disorder falling below pLDDT 50 is nearly identical.